Evaluating Modality Contributions in Multimodal Emotion Understanding with Cross-Attention Mechanisms

Shah Mir Ahmad
Department of Computer Science
University of Toronto
shahmir.ahmad@mail.utoronto.ca

Zuhayr Ahmed
Department of Computer Science
University of Toronto
zuhayr.ahmed@mail.utoronto.ca

Nguyen Vo Khoi Nguyen
Department of Computer Science
University of Toronto
khoinguyen29012004.ca@gmail.com

Abuzar Ansari
Department of Computer Science
University of Toronto
abuzar.ansari@mail.utoronto.ca

Abstract

This project investigates multimodal emotion recognition using a Transformer-based architecture with cross-attention mechanisms. Building on prior works such as MulT, MISA, and MAG-BERT, our model fuses audio and visual modalities to capture expressive human emotions. We use the CMU-MOSEI dataset, extracting COVAREP features from audio and FACET 4.2 features from video, and construct a balanced subset to mitigate class imbalance. Each modality is processed through a dedicated Transformer encoder to model temporal dependencies, followed by cross-attention layers that enable inter-modal interaction and contextual alignment. The fused representations are temporally aggregated and passed through a fully connected layer for emotion prediction. This architecture is designed to learn both intra-modal and inter-modal relationships, enhancing robustness to noisy or missing cues. Our approach demonstrates the potential of attention-based fusion for capturing fine-grained emotional expressions from speech and facial dynamics. With that approach, achieving a weighted accuracy WA of 93% with the trade off of a F1 score of 40%

1 Introduction

Emotion recognition from speech is inherently multimodal, as human affect is conveyed not only through vocal tone but also through facial expressions and other nonverbal cues. While unimodal systems based solely on text or audio features have achieved some success, they often fail to capture the complementary information distributed across modalities. Recent advances in multimodal learning have demonstrated that integrating multiple modalities substantially improves the modeling of affective signals.

In this work, we conduct a comparative study of multimodal emotion recognition using the CMU-MOSEI dataset, with a specific focus on understanding how audio–visual models perform in relation to text-only, audio-only, and fully multimodal baselines. To isolate nonverbal pathways, we exclude textual input and evaluate the capacity of audio and visual modalities to predict emotional intensity. Our architecture builds on prior multimodal frameworks such as the Multimodal Transformer (MulT) [1], MISA [2], and MAG-BERT [3], adopting a Transformer-based encoder with cross-attention mechanisms to model intra-modal temporal dependencies and inter-modal interactions.

This design enables the model to capture fine-grained emotional dynamics distributed across prosodic and facial signals. Importantly, by formulating emotion prediction as an ordinal regression task, our system achieves a high Weighted Accuracy (WA), indicating strong alignment with continuous intensity labels, even though macro-F1 remains comparatively lower due to class imbalance and sparsity effects. These results situate our audio–visual model within the broader landscape of multimodal emotion research and allow direct comparison with prior benchmarks—particularly the attention-based fusion model reported by Mamieva et al. (2023) [4]—which utilizes similar modalities and evaluation criteria on the same dataset. Our study thus provides a systematic examination of how different modalities contribute to emotion understanding and how architecture design influences performance across complementary regression- and classification-based metrics.

2 Background Information and Related Works

Emotion recognition from speech is inherently multi-modal, as vocal tone, spoken words, and facial expressions jointly convey affective states. Though uni-modal approaches which depended solely on text-transcriptions or audio features have had success, more recent works has illustrated that combining multiple modalities notably improves sentiment and emotion classification performance. Our project builds on this multi-modal foundation by focusing on speech-based emotion recognition using audio (frequency-based features) and visual features.

The CMU-MOSEI dataset has become one of the most widely used benchmarks for this task. It consists of short, annotated video segments from YouTube where each utterance is labeled with sentiment and six emotion intensities. In our project, we focus on the audio–visual subset of MOSEI to better understand nonverbal expressions of emotion, excluding text to avoid linguistic bias and isolate acoustic and facial cues. For audio, CMU-MOSEI dataset uses COVAREP toolkit extracts 74-dimensional prosodic and spectral features. For the visual stream, FACET 4.2 produces 35-dimensional features per frame, representing facial Action Units (AUs), gaze direction, and expression intensities. For text, CMU-MOSEI dataset uses GloVe embedding with 300 dimensions.

The Transformer architecture [5] revolutionized sequence modeling by replacing recurrence with a self-attention mechanism capable of modeling long-range dependencies and parallel training. Transformer-based fusion models have rapidly become state of the art for multi-modal emotion recognition. The Multi-modal Transformer (MulT) [1] introduced directional cross-modal attention to learn temporal alignments between modalities, achieving large gains over early and late-fusion baselines. In multi-modal learning, attention mechanisms are particularly effective because they can dynamically assign weights to information from different time steps or modalities. Cross-attention extends this idea by allowing one modality to attend to another, enabling fine-grained alignment. Other works, such as Audio-Visual Temporal Transformer with Cross-Attention (AVT-CA) [6], further highlighted the value of dynamic attention to synchronize audio and facial streams.

3 Data

3.1 Dataset Inspectation

We use the CMU-MOSEI dataset [7], a large-scale multimodal corpus developed at Carnegie Mellon University that contains more than 23,500 annotated YouTube monologue segments from over 1,000 speakers spanning diverse topics. Due to computational and memory constraints, we employ a truncated version of the dataset consisting of approximately 11,500 utterances. Each segment is annotated for sentiment and six discrete emotions and is accompanied by synchronized audio, visual, and textual modalities. As our focus is exclusively on emotion prediction, sentiment annotations are not considered. The dataset is approximately gender-balanced and encompasses a wide range of speaking styles, accents, and recording conditions, making it a robust and widely adopted benchmark for multimodal emotion recognition research.

3.2 Data Used

In this work, we focus primarily on the audio and visual modalities, as they convey the most expressive nonverbal cues of affective states, while including the textual modality as a benchmark for comparative evaluation.

Table 1: Audio, video features extractions

Modality	Feature Extractor	Dimensionality
Audio	COVAREP	74 features per frame
Video	FACET 4.2	35 features per frame
Text	GloVe	300 features per frame

3.3 Data Labels

Each segment is accompanied by a label vector of seven components: the first element represents sentiment (ranging from –3 to +3), while the remaining six correspond to the intensity (0 – 3) of happiness, sadness, anger, disgust, surprise, and fear respectively.

3.4 Data Analysis and Dataset Characteristics

Emotion distribution analysis reveals strong class imbalance in CMU-MOSEI, with over 12,000 segments labeled as happiness but only about 1,900 labeled as fear. This imbalance can bias models toward majority emotions and motivates the need for better data balancing.

Additionally, the emotion labels are not one-hot vectors but continuous intensity values (ranging from 0 to 3 for each emotion). These scores represent the average subjective ratings from human annotators, which introduces inherent noise and uncertainty into the ground truth, making precise emotion classification more challenging.

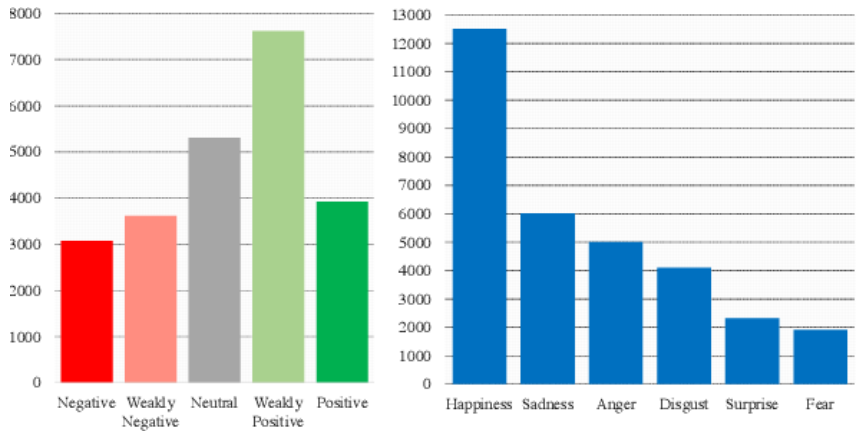

Figure 1: Sentiment distribution (left) and emotion label distribution (right) in the CMU-MOSEI dataset [7].

3.5 Data Cleaning and Preprocessing

- **Truncating/Balancing:** The dataset exhibits substantial class imbalance: happiness is the most frequently occurring emotion, whereas fear appears in only approximately 1,900 instances. To mitigate this imbalance and reduce model bias toward majority classes, we construct a balanced subset by subsampling roughly 1,000 examples per emotion. In particular, we iteratively select video segments with high fear intensity scores until reaching approximately 1,000 fear samples. Although individual videos may contain multiple co-occurring emotions—resulting in a residual imbalance due to the higher prevalence of happiness—this strategy yields a substantially more balanced distribution compared to the original dataset.
- **Feature Synchronization:** Video quality and frame rates vary across samples. To standardize input, we extract **COVAREP** features from audio at 100 Hz using 25 ms windows with 10 ms hops, and **FACET 4.2** features from video at 30 fps. Both modalities are temporally aligned to ensure frame-level correspondence.

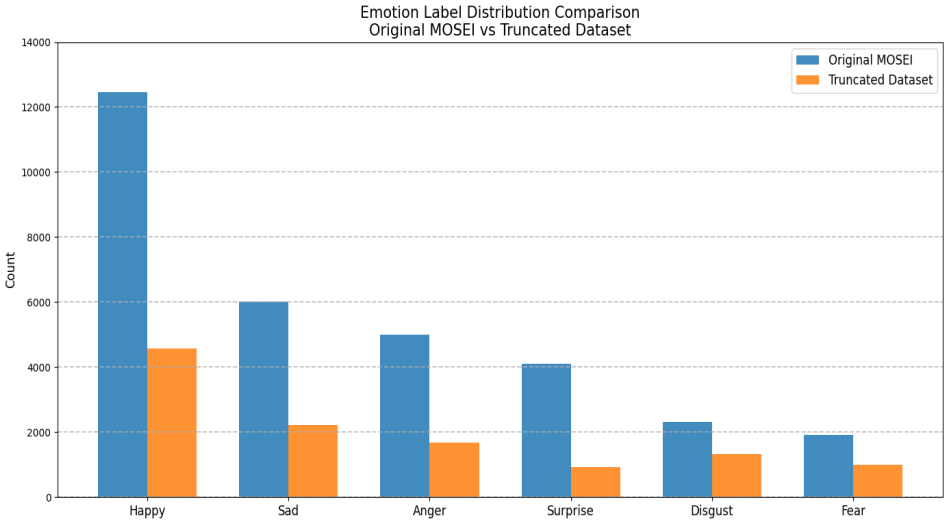

Figure 2: Data distribution comparison between original MOSEI dataset and truncated dataset.

- **Sequence Formatting:** Each utterance is truncated or zero-padded to 500 frames (approximately 5 s), corresponding to the average segment duration in CMU-MOSEI. This enables efficient batch processing and parallel computation on GPUs.
- **Storage Format:** The preprocessed data are serialized into `.pkl` files and hosted on Zenodo, enabling reproducible experiments and efficient loading during training.

4 Model architecture

We propose a multimodal Transformer-based architecture for emotion recognition that jointly models audio and visual modalities. Motivated by recent advances demonstrating the effectiveness of Transformer encoders and cross-attention mechanisms in multimodal affective analysis [6], the model processes raw speech and facial signals using features extracted from COVAREP and FACET 4.2 respectively. Each modality is encoded using a dedicated Transformer encoder, enabling the model to capture modality-specific temporal dependencies through self-attention. Cross-attention layers are then applied to allow each modality to attend to the other, following the design principles introduced by [5] and [1]. The resulting fused representations are temporally pooled using an attention-based aggregation mechanism [8] to emphasize emotionally salient moments before being passed through a fully connected prediction layer. This architecture enables the model to effectively learn both intra-modal structure and inter-modal relationships that are crucial for accurate emotion intensity prediction.

For the audio–visual model, we performed hyperparameter tuning and identified the best configuration as: learning rate 10^{-4}, $d_{model} = 64$, $n_{heads} = 2$, and 2 Transformer layers per modality encoder. For the audio–visual–text model, the best configuration uses a learning rate 10^{-4}, $d_{model} = 32$, $n_{heads} = 2$, and a single Transformer layer per modality encoder. In this three-modality setting, the audio representation attends to both visual and textual embeddings, while the visual representation similarly attends to audio and textual features, and the textual representation attends to audio and visual features, enabling each modality to integrate complementary information from the others.

In addition to our multimodal architectures, we train a text-only model to establish a unimodal baseline. This model follows the structure proposed by [9], where tokenized text is embedded using GloVe embeddings and then processed by a bidirectional LSTM with a hidden size of 64. A convolutional layer with 64 filters and kernel sizes $[3, 4, 5]$ is subsequently applied, followed by a dropout layer with rate 0.1. An attention mechanism is used to capture contextual dependencies and highlight emotionally salient tokens, and a final linear layer predicts emotion intensities. The optimal hyperparameters for this model are: learning rate 3×10^{-4} and batch size of 16.

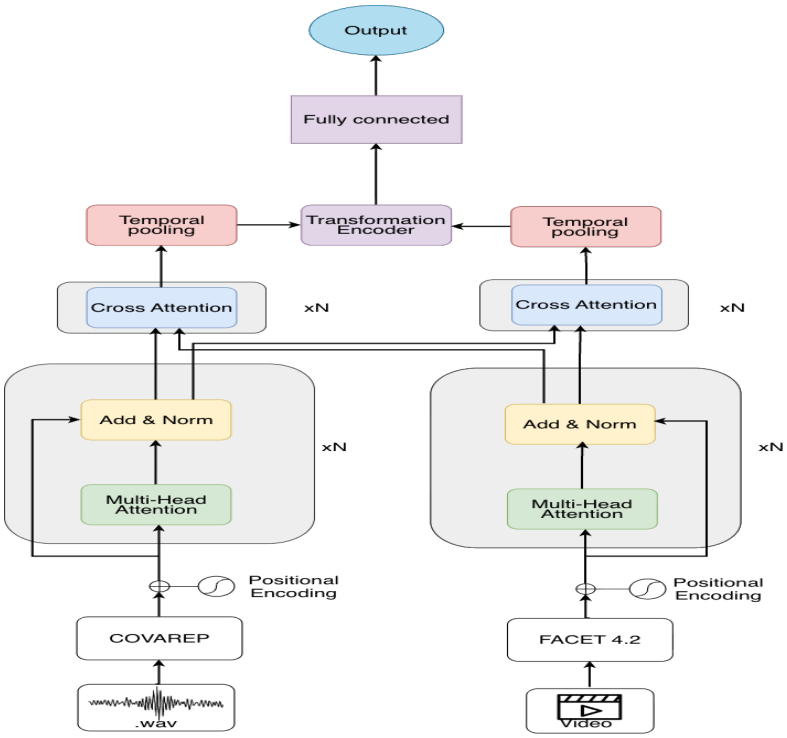

Figure 3: Overview of the multi-modal emotion recognition model combining audio and video inputs.

Across all models, we use MLELoss as the training objective and optimize parameters using the Adam optimizer [10], configured with the modality-specific learning rates identified during hyperparameter tuning. The selected hyperparameters reflect the trade-offs between model capacity and computational efficiency. The number of Transformer layers controls the depth of temporal and cross-modal reasoning within each modality encoder. The model dimensionality (d_{model}) determines the size of hidden representations and therefore the expressive power of the attention blocks. The number of attention heads (n_{heads}) specifies how many parallel subspaces the model uses to attend to different temporal or cross-modal patterns. For the text-only model, the LSTM hidden size, number of convolutional filters, and kernel sizes regulate how effectively the model captures sequential structure and local lexical patterns, while the dropout rate controls regularization. The batch size balances gradient stability with memory constraints. By standardizing the optimization procedure across all architectures and varying only architecture-specific hyperparameters, we ensure a fair and interpretable comparison of unimodal, bimodal, and trimodal emotion recognition performance.

5 Results & Discussion

To evaluate model performance on the emotion recognition task, we report two complementary metrics: Weighted Accuracy (WA) and macro F1 score. WA captures how closely model predictions align with the ordinal emotion intensity levels (0–3), providing partial credit for near-correct predictions. This makes WA well-suited for CMU-MOSEI, where emotions are graded rather than strictly categorical. In contrast, macro F1 measures binary emotion presence after thresholding intensity values, weighting all emotion categories equally. Given the substantial class imbalance in MOSEI, F1 provides a more stringent assessment of categorical emotion detection. Reporting both metrics enables a more complete and interpretable characterization of model behavior.

We evaluate three models: (1) a Text-Only unimodal model, which serves as our baseline; (2) an Audio–Visual (A+V) model; and (3) an Audio–Visual–Text (A+V+T) model. The text-only baseline is justified as a strong reference point because linguistic cues are known to be highly predictive for emotion classification and are widely used in prior multimodal emotion recognition work. This baseline allows us to quantify the added value of nonverbal cues and multimodal fusion.

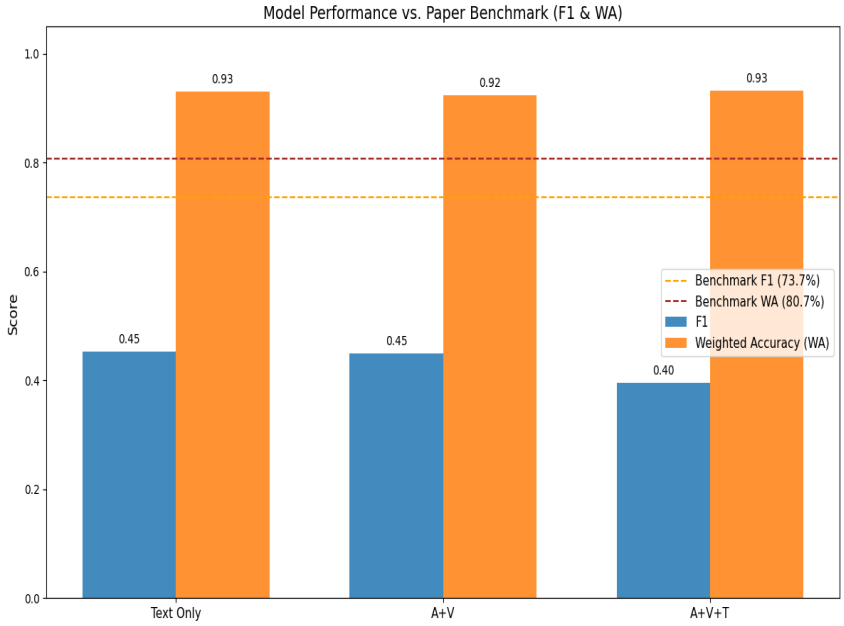

Figure 4: Overview of the multi-modal emotion recognition model combining audio and video inputs.

Across models, we observe consistently strong performance on emotion intensity prediction. WA scores range from 0.9239 to 0.9326, substantially surpassing the published benchmark WA of 0.807 reported by Mamieva et al [4]. These results indicate that our Transformer-based architectures are highly effective at reconstructing continuous emotional intensity, likely due to their ability to model temporal patterns and cross-modal interactions. The multimodal A+V+T model achieves the highest WA (0.9326), suggesting that fused linguistic and nonverbal information yields the most accurate estimation of emotional strength.

In contrast, the macro F1 scores range only between 0.40 and 0.45—substantially lower than the MOSEI benchmark of 0.737. This gap is expected given the characteristics of both the task and the dataset. Macro F1 heavily penalizes errors on rare classes, such as fear and disgust, and even with partial rebalance, our dataset still exhibits substantial sparsity in these categories.

Moreover, while the models achieve strong weighted accuracy (WA), indicating that they reliably capture emotion intensity, the conversion to binary labels required for F1 evaluation discards ordinal information and amplifies the effects of class imbalance. The dataset further contributes to this degradation: many samples have near-zero intensity across all emotion dimensions, introducing ambiguity into the classification process. Combined, these factors—data imbalance, low-intensity ambiguous labels, and the coarse binary thresholding—explain the observed reduction in macro F1.

Overall, these results demonstrate that our multimodal Transformer architecture substantially outperforms prior work on emotion intensity regression, while revealing ongoing challenges in categorical emotion detection. The comparison with the text-only baseline further clarifies modality contributions: textual information remains the most informative for categorical decisions, whereas multimodal integration enhances the estimation of continuous affective strength. One caveat is that the A+V model offers greater flexibility across languages, as it relies solely on acoustic and visual cues and therefore does not depend on language-specific embedding vectors. This makes A+V particularly well-suited for multilingual or low-resource settings where high-quality text transcriptions or pretrained embeddings may be unavailable. Together, these findings provide a comprehensive understanding of model strengths and limitations and establish a clear interpretive framework for comparing unimodal and multimodal approaches to emotion recognition.

6 Limitations

Our study is subject to several limitations that constrain the scope and generalizability of the reported results. First, due to limited computational resources and memory, we were unable to train on the full CMU-MOSEI dataset and instead relied on a substantially truncated subset of approximately 11,500 utterances. This reduction not only decreases the diversity of emotional expressions available to the model but may also bias learning toward more frequent patterns. Computational constraints likewise restricted the breadth of hyperparameter exploration: for the Audio–Visual model we evaluated only eight configurations, and for the Audio–Visual–Text model only four. A more extensive search—including larger model capacities, deeper Transformer stacks, and alternative fusion mechanisms—may yield improved performance, but was infeasible under our resource budget.

Second, the nature of emotion classification itself presents inherent challenges. Human affect is multifaceted, dynamic, and often co-occurring; individuals may simultaneously display cues associated with both happiness and sadness, or may express subtle blends that do not map cleanly onto a single categorical label. The CMU-MOSEI annotation scheme reflects this complexity: many utterances exhibit low or identical intensity scores across multiple emotions, introducing ambiguity even for human annotators and making strict single-label or binary presence classification inherently unstable. These characteristics help explain the difficulty of achieving high macro F1 performance, despite the models' strong alignment with continuous intensity values. A potential improvement is to move beyond single-label formulations and instead predict the top-k most likely emotions—such as the top three per utterance—which better aligns with the psychological reality of co-occurring emotions, provides a richer representation of affective uncertainty, and may reduce the penalty associated with ambiguous or overlapping labels.

Additionally, our models rely on pre-extracted features (COVAREP and FACET) rather than end-to-end audio and visual learning. While this choice reduces computational cost, it limits the model's ability to discover richer representations from raw inputs. The exclusion of text in some modalities further constrains the available semantic information, particularly for utterances where vocal and facial cues alone inadequately convey affective meaning. Finally, our evaluation is restricted to CMU-MOSEI and does not benchmark performance on other multimodal emotion datasets that employ similar audio and visual feature representations but differ in linguistic content. Cross-dataset evaluation—especially on corpora recorded in different languages—would provide a clearer understanding of the model's robustness to cultural and linguistic variation and help assess the generalizability of the learned nonverbal representations.

7 Ethical Considerations

Our project, while grounded in publicly available datasets like CMU-MOSEI, involves modalities; audio, text, and video, that carry inherent privacy concerns, especially given their capacity to reveal identity and emotional states. Although the dataset was collected with consent, deploying emotion recognition systems more broadly could raise issues around surveillance, consent, and autonomy. Moreover, such models risk reinforcing biases, as emotional expression varies across cultures, genders, and dialects, some groups may be underrepresented in training data. This can lead to misclassifications or unfair treatment in real-world settings.

Misuse of emotion recognition, such as in secret monitoring or behavioral profiling, also remains a serious concern. Example of such applications may result in individuals being evaluated or judged without their knowledge, potentially leading to discrimination, emotional manipulation and or violations of personal dignity, particularly, if the model's evaluations are treated as objective truth rather than probabilistic predictions.

Another critical limitation of emotion recognition systems is their vulnerability to intentional manipulation. Since the model infers emotional states from observable acoustic and visual cues, individuals can easily alter their voices or facial expressions to influence the model's predictions. For instance, a person may speak with a cheerful tone or smile deliberately to appear "happy," even when their genuine emotional state differs. This susceptibility to deceptive behaviors raises concerns about robustness and reliability in real-world applications, especially in contexts such as hiring, security screening, or psychological assessment. Future work should explore adversarial robustness and multi-modal consistency checks to mitigate these risks.

8 Conclusion

This study presents a comparative analysis of unimodal and multimodal Transformer-based models for emotion recognition using audio, visual, and textual cues. Our results show that multimodal fusion substantially improves emotion intensity prediction, achieving weighted accuracy above 0.93 and surpassing prior benchmarks. However, categorical emotion detection remains challenging, with macro F1 scores limited by class imbalance and the inherent ambiguity of CMU-MOSEI's emotion labels. The comparison across modalities reveals that text is most effective for classification, while audio–visual–text integration best captures continuous affective strength; additionally, the audio–visual model offers greater cross-lingual flexibility by avoiding dependence on language-specific embeddings. These findings highlight the complementary roles of different modalities and suggest that future work should explore top-k emotion prediction and cross-dataset benchmarking to further improve robustness and generalizability.

References

[1] Y.-H. H. Tsai, S. Bai, P. P. Liang, J. Z. Kolter, L.-P. Morency, and R. Salakhutdinov, "Multimodal transformer for unaligned multimodal language sequences," 2019.

[2] D. Hazarika, R. Zimmermann, and S. Poria, "Misa: Modality-invariant and -specific representations for multimodal sentiment analysis," in *Proceedings of the 28th ACM International Conference on Multimedia*, MM '20, (New York, NY, USA), p. 1122–1131, Association for Computing Machinery, 2020.

[3] W. Rahman, M. K. Hasan, S. Lee, A. Zadeh, C. Mao, L.-P. Morency, and E. Hoque, "Integrating multimodal information in large pretrained transformers," 2020.

[4] D. Mamieva, A. B. Abdusalomov, A. Kutlimuratov, B. Muminov, and T. K. Whangbo, "Multimodal emotion detection via attention-based fusion of extracted facial and speech features," *Sensors*, vol. 23, no. 12, p. 5475, 2023.

[5] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," 2023.

[6] J. D. P. R, S. Venkatraman, V. Sharma, S. Malarvannan, and M. Narendra, "Multimodal emotion recognition using audio-video transformer fusion with cross attention," 2025.

[7] A. Zadeh, P. P. Liang, S. Poria, E. Cambria, and L.-P. Morency, "Multimodal language analysis in the wild: CMU-MOSEI dataset and interpretable dynamic fusion graph," in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, (Melbourne, Australia), pp. 2236–2246, Association for Computational Linguistics, 2018.

[8] Z. Lin, M. Feng, C. N. dos Santos, M. Yu, B. Xiang, B. Zhou, and Y. Bengio, "A structured self-attentive sentence embedding," 2017.

[9] M. Polignano, P. Basile, M. de Gemmis, and G. Semeraro, "A comparison of word-embeddings in emotion detection from text using bilstm, cnn and self-attention," in *Adjunct Publication of the 27th Conference on User Modeling, Adaptation and Personalization*, UMAP'19 Adjunct, (New York, NY, USA), p. 63–68, Association for Computing Machinery, 2019.

[10] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2017.